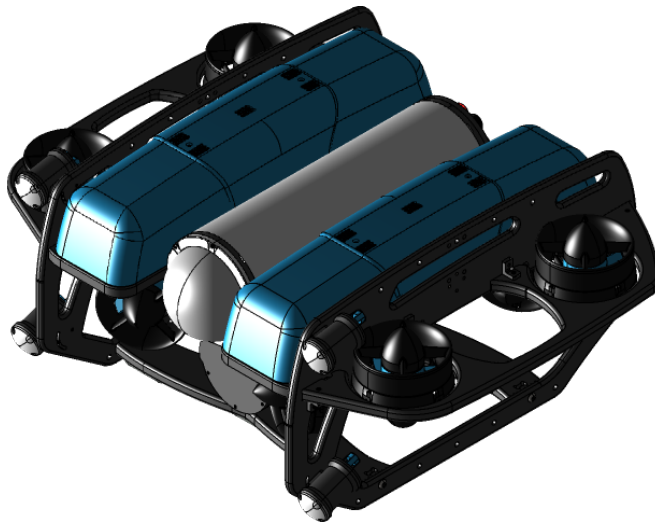


Field and Service Robotics Technical Project

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1 Introduction

This project explores the modeling and control of underwater robots, with a focus on the BlueROV2 Heavy, an eight-thruster Remotely Operated Vehicle (ROV). The study develops a 6-DOF dynamic model, incorporating hydrodynamic effects such as added mass, damping, and buoyancy, while using quaternions to avoid orientation singularities. Two nonlinear control strategies are implemented: a sliding mode controller and a feedback linearization approach. The goal is to evaluate their performance in simulation, ensuring precise and stable underwater navigation. Underwater robots play a crucial role in exploration, inspection, and environmental monitoring, making reliable control systems essential for real-world applications.

2 Reference Frames

When modeling an Underwater Robot two reference frames are needed to describe the motion

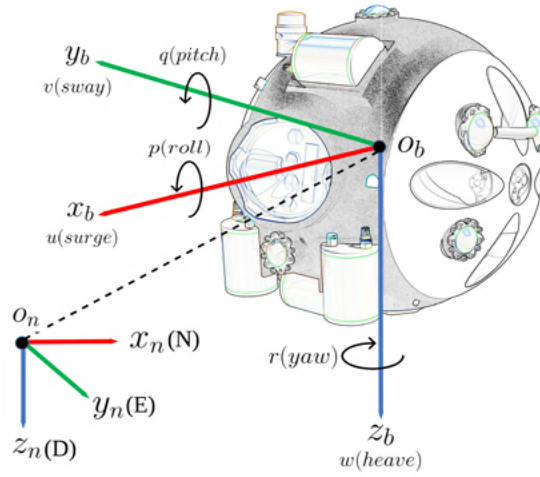


Figure 1: Reference Frames used in the equations of motion

- North-East-Down (NED) reference frame, with orthonormal basis x_n, y_n, z_n and origin o_n . The NED world frame refers to the real world in which the x_n, y_n, z_n axes point towards north, east and downwards normal to the Earth's surface, respectively.
- Body reference frame with axes x_b, y_b, z_b and origin o_b . The body frame of an ROV is a moving coordinate frame that is fixed to the vehicle. The

origin is generally defined at the geometric centre of the vehicle in order to exploit physical symmetries.

3 Nomenclature

The motion of an Underwater Robot in 6 DoFs can be represented in a vectorial form using the SNAME notation (see Table 1) where six individual coordinates are generalised to describe the position and orientation and the linear and angular velocities of the vehicle. According to this notation the generalised pose

No.	DoF	Force/Moment	Velocity	Pos./Angle
1	Surge	X	u	x
2	Sway	Y	v	y
3	Heave	Z	w	z
4	Roll	K	p	ϕ
5	Pitch	M	q	θ
6	Yaw	N	r	ψ

Table 1: SNAME notation for marine vessels

and velocity coordinates can be addressed by the following vectors:

$$\eta = [x \ y \ z \ \phi \ \theta \ \psi]^T \quad (1)$$

$$\nu = [u \ v \ w \ p \ q \ r]^T \quad (2)$$

where $\eta \in \mathbb{R}^6$ denotes the position and orientation vector in the NED coordinate system, and $\nu \in \mathbb{R}^6$ denotes the linear and angular velocity vector in the body-fixed reference frame. Moreover we can describe the forces and moments acting on the vehicle in the body-fixed reference frame *via* the vector:

$$\tau = [X \ Y \ Z \ K \ M \ N]^T \in \mathbb{R}^6 \quad (3)$$

4 Quaternion Representation

In order to avoid the singularity of the Euler angles the representation with unitary quaternions was adopted. The notation used for the quaternion is the following

$$\mathbf{q} = [q \ \epsilon_1 \ \epsilon_2 \ \epsilon_3] = [q \ \epsilon] \quad (4)$$

where q represents the scalar part of the quaternion, whereas ϵ it's the vector part. The use of unit quaternions implies that the dimension of the vector η changes accordingly since now the pose is given by

$$\eta = [x \ y \ z \ q \ \epsilon_1 \ \epsilon_2 \ \epsilon_3] \in \mathbb{R}^7 \quad (5)$$

As we will see, this choice also implies that the Jacobian will no longer be a square matrix, which will change the formulation of the control laws used,

while the underlying logic remains unchanged. Finally it's worth recalling how the attitude error is computed when using quaternion representation; given the quaternion $Q_d = \{q_d \quad \epsilon_d\}$ describing the desired orientation of the vehicle in the NED frame, and $Q = \{q \quad \epsilon\}$ the quaternion related to the actual orientation of the body in the body frame, then the error is expressed by the quaternion $\tilde{Q} = Q^{-1} * Q_d = \{\tilde{\eta} \quad \tilde{\epsilon}\}$, where $\tilde{\eta}$ and $\tilde{\epsilon}$ are given by¹:

$$\tilde{\epsilon} = \eta \epsilon_d - \eta_d \epsilon + \mathbf{S}(\epsilon_d) \epsilon \quad (6)$$

$$\tilde{\eta} = \eta \eta_d + \epsilon^T \epsilon_d \quad (7)$$

5 Kinematic Model

The kinematic model is given by

$$\dot{\eta} = \mathbf{J}_q(\mathbf{q}) \nu \quad (8)$$

Hence, the jacobian matrix from the vehicle body frame to the NED world reference frame using unit quaternion representation is given by:

$$\mathbf{J}_q(\eta) = \begin{bmatrix} \mathbf{R}_b^n(\mathbf{q}) & \mathbf{0}_{3 \times 3} \\ \mathbf{0}_{4 \times 3} & \mathbf{T}_q(\mathbf{q}) \end{bmatrix} \quad (9)$$

where the matrix $\mathbf{R}_b^n(\mathbf{q}) \in \mathbb{R}^{3 \times 3}$ represents the rotation from the body frame to the inertial (NED) frame, and is computed from the unit quaternion $\mathbf{q} = [q, \epsilon_1, \epsilon_2, \epsilon_3]^T$ as:

$$\mathbf{R}_b^n(\mathbf{q}) = \begin{bmatrix} 1 - 2(\epsilon_2^2 + \epsilon_3^2) & 2(\epsilon_1 \epsilon_2 - \epsilon_3 q) & 2(\epsilon_1 \epsilon_3 + \epsilon_2 q) \\ 2(\epsilon_1 \epsilon_2 + \epsilon_3 q) & 1 - 2(\epsilon_1^2 + \epsilon_3^2) & 2(\epsilon_2 \epsilon_3 - \epsilon_1 q) \\ 2(\epsilon_1 \epsilon_3 - \epsilon_2 q) & 2(\epsilon_2 \epsilon_3 + \epsilon_1 q) & 1 - 2(\epsilon_1^2 + \epsilon_2^2) \end{bmatrix} \quad (10)$$

The matrix $\mathbf{T}_q(\mathbf{q}) \in \mathbb{R}^{4 \times 3}$ relates the angular velocity vector expressed in the body frame to the time derivative of the quaternion orientation. It is defined as:

$$\mathbf{T}_q(\mathbf{q}) = \frac{1}{2} \begin{bmatrix} -\epsilon_1 & -\epsilon_2 & -\epsilon_3 \\ q & -\epsilon_3 & \epsilon_2 \\ \epsilon_3 & q & -\epsilon_1 \\ -\epsilon_2 & \epsilon_1 & q \end{bmatrix} \quad (11)$$

6 Dynamic Model

The 6-DOF dynamic model of an Underwater robot is given by the non linear equations:

$$\mathbf{M} \dot{\nu} + \mathbf{C}(\nu) \nu + \mathbf{D}(\nu) \nu + \mathbf{g}(\eta) = \mathbf{B} \tau \quad (12)$$

with

¹ $S(\cdot)$ is the skew-symmetric operator



Figure 2: BlueROV2 Heavy

$$\mathbf{M} = \mathbf{M}_A + \mathbf{M}_{RB} > 0$$

$$\mathbf{C}(\nu) = \mathbf{C}_{RB}(\nu) + \mathbf{C}_A(\nu)$$

$$\mathbf{D}(\nu) = \mathbf{D}_L + \mathbf{D}_{NL}(\nu)$$

Where $\mathbf{B}$ is used as a mapping matrix for the thruster configuration, and $\mathbf{g}(\cdot)$ is the vector of hydrostatic forces and moments due to gravitational and buoyant effects acting on the vehicle. The system inertia matrix $\mathbf{M}$ is a positive semi-definite matrix composed of the rigid-body inertia matrix $\mathbf{M}_{RB}$ and the added-mass (hydrodynamic inertia) matrix $\mathbf{M}_A$. The Coriolis and centripetal matrix $\mathbf{C}(\cdot)$ consists of the rigid-body Coriolis matrix $\mathbf{C}_{RB}(\cdot)$ and the added-mass Coriolis matrix $\mathbf{C}_A(\cdot)$. Finally, the total hydrodynamic damping matrix $\mathbf{D}(\cdot)$ is the sum of the linear damping matrix $\mathbf{D}_L$ and the nonlinear damping matrix $\mathbf{D}_{NL}(\cdot)$. For the analytical expressions of the dynamic terms comparing in the model see [1] and [2].

7 BlueROV2

The physical system used for this project is the BlueROV2 Heavy (BlueRobotics 2018a), depicted in Fig.2. This BlueROV in Heavy configuration has eight

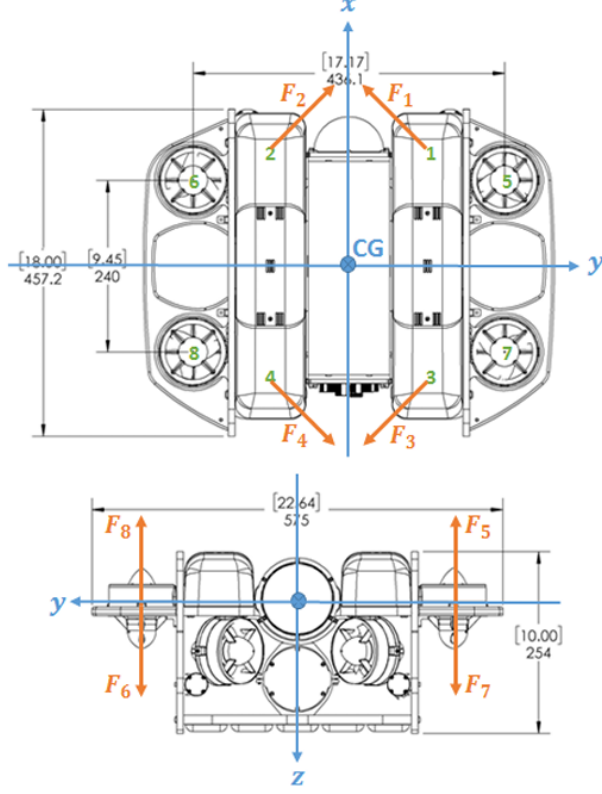


Figure 3: Thrusters Configuration

thrusters instead of six. The thrust configuration matrix $B \in \mathbb{R}^{6 \times 8}$ that appears in equation 12, after some computations (see [1]), is given by

$$B = \begin{bmatrix} 0.707 & 0.707 & -0.707 & -0.707 & 0 & 0 & 0 & 0 \\ -0.707 & 0.707 & -0.707 & 0.707 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 1 & 1 & -1 \\ 0.06 & -0.06 & 0.06 & -0.06 & -0.218 & -0.218 & 0.218 & 0.218 \\ 0.06 & 0.06 & -0.06 & -0.06 & 0.120 & -0.120 & 0.120 & -0.120 \\ -0.1888 & 0.1888 & 0.1888 & -0.1888 & 0 & 0 & 0 & 0 \end{bmatrix} \quad (13)$$

which is a full-rank matrix (i.e. $\rho(B) = 6$). Since the system has 6-DOF (Fig. 3) we can infer that such a system is fully-actuated (from an heuristic point of view this is also clear if one looks at the thrusters configuration).

For simplicity's sake, the allocation matrix was not taken into account in the model simulation, and the vector τ of generalized forces and moments was entered directly into the system. Finally the physical parameters of the ROV are summarized in Table-2.

Parameter	Value	Unit
m	11.5	kg
W	112.8	N
B	114.8	N
$\mathbf{I}$	diag(0.16,0.16,0.16)	kg·m ²
$X_{\dot{u}}$	-5.5	kg
$Y_{\dot{v}}$	-12.7	kg
$Z_{\dot{w}}$	-14.57	kg
$K_{\dot{p}}$	-0.12	kg·m ² /rad
$M_{\dot{q}}$	-0.12	kg·m ² /rad
$N_{\dot{r}}$	-0.12	kg·m ² /rad
X_u	-4.03	Ns/m
Y_v	-6.22	Ns/m
Z_w	-5.18	Ns/m
K_p	-0.07	Ns/rad
M_q	-0.07	Ns/rad
N_r	-0.07	Ns/rad
$X_{u u }$	-18.18	Ns ² /m ²
$Y_{v v }$	-21.66	Ns ² /m ²
$Z_{w w }$	-36.99	Ns ² /m ²
$K_{p p }$	-1.55	Ns ² /rad ²
$M_{q q }$	-1.55	Ns ² /rad ²
$N_{r r }$	-1.55	Ns ² /rad ²

Table 2: Physical parameters of the BlueROV2

8 Sliding Mode Controller

The idea behind the sliding controller starts from the definition of the sliding surface

$$\mathbf{s} = \dot{\tilde{\mathbf{x}}} - \Lambda \tilde{\mathbf{x}} = 0 \quad (14)$$

where $\tilde{\mathbf{x}} = \mathbf{x}_d - \mathbf{x}$ is the state error, $\mathbf{x}$ is the state of the system and Λ is a positive definite matrix. When the sliding condition is satisfied the state converges (*slide*) toward the desired value $\mathbf{x}_d$. Then the control input is designed with the aim of force the state to stay on the sliding surface. This fact implies an high control activity and the typical *chattering* behaviour. The designed control law is the following

$$\mathbf{u} = \mathbf{B}^\dagger [\mathbf{K}_D \mathbf{s} + \mathbf{g}(\boldsymbol{\eta}) + \mathbf{K}_S \text{sgn}^*(\mathbf{s})] \quad (15)$$

where $\mathbf{B}^\dagger$ is the pseudo-inverse of the thruster allocation matrix², $\mathbf{K}_D, \mathbf{K}_S$ are positive definite matrices, $\mathbf{g}(\boldsymbol{\eta})$ is the hydrostatic forces vector and $\text{sgn}^*(\mathbf{s})$ is

²Since the project has been carried out only in simulation we will assume $\mathbf{B}$ as an identity matrix.

not the typical *signum* function, since that choice causes the presence of high-frequency components in the input (*chattering*); to limit this effect the j -th component of the function has been computed as

$$\text{sgn}^*(s)_j = \begin{cases} 1 & \text{if } s > \epsilon \\ -1 & \text{if } s \leq -\epsilon \\ \frac{s_j}{\epsilon} & \text{otherwise} \end{cases} \quad (16)$$

where ϵ represents the amplitude of the so called *boundary layer*: by choosing a small value of ϵ we will experience a more noticeable effect of chattering, but the norm of the error will be smaller. The proof of asymptotic stability of the control law can be found in [3].

9 Feedback Linearization Controller

The main principle with feedback linearization is the transformation of the non linear dynamics of the system into a set of independent chain of integrators, as follows

$$\ddot{\eta} = \lambda^n \quad (17)$$

$$\dot{\nu} = \lambda^b \quad (18)$$

where λ^n can be viewed as the commanded accelerations in the NED frame, while λ^b can be seen as the body-fixed commanded acceleration vector. Let's consider the kinematic and dynamic models

$$\dot{\eta} = J_q(\mathbf{q})\nu \quad (19)$$

$$M\dot{\nu} + \mathbf{n}(\boldsymbol{\eta}, \boldsymbol{\nu}) = \boldsymbol{\tau} \quad (20)$$

where $\mathbf{n}(\boldsymbol{\eta}, \boldsymbol{\nu}) = C(\boldsymbol{\nu})\boldsymbol{\nu} + D(\boldsymbol{\nu})\boldsymbol{\nu} + \mathbf{g}(\boldsymbol{\eta})$. We consider the control law

$$\boldsymbol{\tau} = \mathbf{n}(\boldsymbol{\eta}, \boldsymbol{\nu}) + M\boldsymbol{\lambda}^b \quad (21)$$

The commanded acceleration in the body frame is chosen as

$$\boldsymbol{\lambda}^b = J_q(\mathbf{q})^\dagger [\boldsymbol{\lambda}^n - \dot{J}_q(\mathbf{q}, \dot{\mathbf{q}})\boldsymbol{\nu}] \quad (22)$$

where $J_q(\mathbf{q})^\dagger$ is the pseudo-inverse of the Jacobian (9), in fact recall that $J_q \in \mathbb{R}^{7 \times 6}$, since we are adopting Unit Quaternion to represent orientation. The commanded acceleration in the NED frame is computed as a PD control law with a feedforward in acceleration

$$\boldsymbol{\lambda}^n = \ddot{\boldsymbol{\eta}}_d + K_D \dot{\tilde{\boldsymbol{\eta}}} + K_P \tilde{\boldsymbol{\eta}} \quad (23)$$

It can be shown that, with simple analytical steps, by selecting this control law results in the linear decoupled system as desired (see [2]). The positive definite matrices K_D and K_P can be tuned in order to set the desired behaviour of the tracking error $\tilde{\boldsymbol{\eta}} = \boldsymbol{\eta}_d - \boldsymbol{\eta}$, whose dynamic is described by the equation $\ddot{\tilde{\boldsymbol{\eta}}} + K_D \dot{\tilde{\boldsymbol{\eta}}} + K_P \tilde{\boldsymbol{\eta}} = 0$.

10 Results and Comparison

The simulation has been carried out in Matlab-Simulink³. The initial condition is

$$\eta_0 = [3.5 \quad 0 \quad 9.5 \quad 1 \quad 0 \quad 0 \quad 0]^T$$

For the positional part the following trajectory was assigned

$$x(t) = 4\cos(0.8t)$$

$$y(t) = 4\sin(0.8t)$$

$$z(t) = 10 + 0.6t$$

while for the orientational part it was only required to remain constantly to the initial value

10.1 Simulation Sliding Mode

The gains of the control law 15 where chosen as

- $K_D = 8$
- $K_S = 67$
- $\Lambda = 58$

This values have been obtained *via* trial and error, until the position and rotation errors were below a certain treshold. It was noticed that the parameters that influenced the most the tracking precision were K_D and Λ . In Fig. 4-5 we can observe the position and orientation errors and that we have obtained norm-bounded errors.

³Code and Media available at [github/salvatore.deluca](https://github.com/salvatore.deluca)

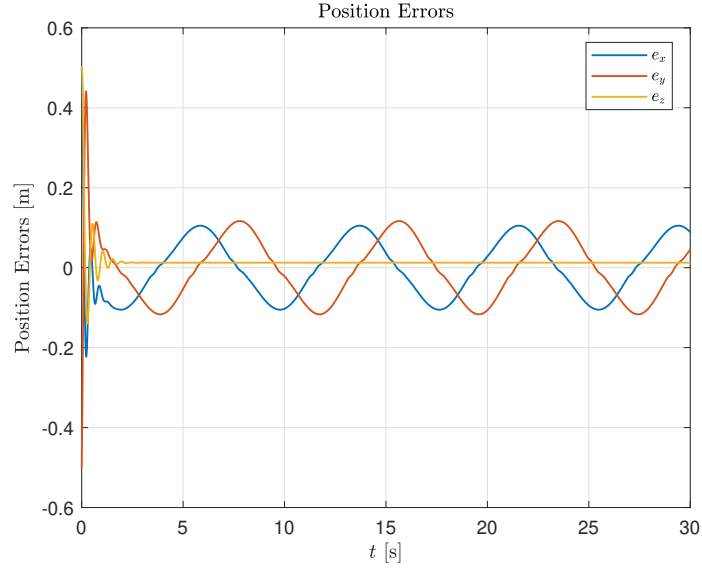


Figure 4: Positions Errors Sliding Mode Control

From Fig.6 we can notice that the forces are bounded, except for the initial spikes caused by the initial conditions, and that the x and y components are the same but simply shifted one with respect to another. Same observation can be made looking at the torques in Fig.7.

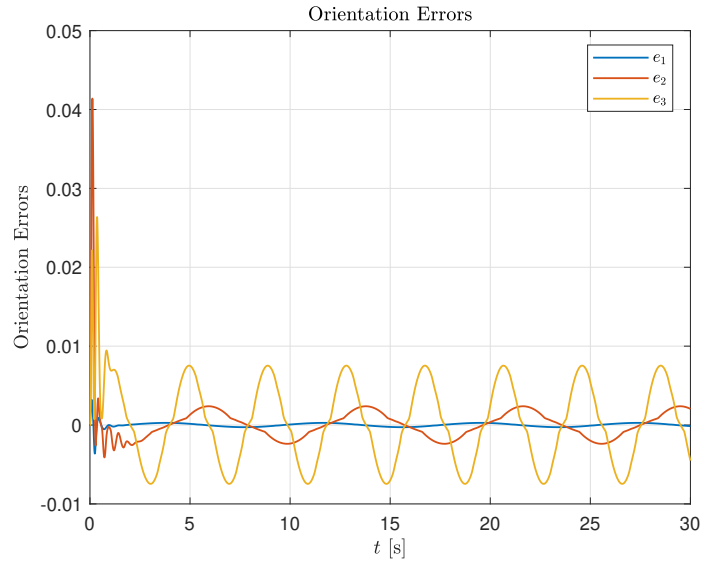


Figure 5: Orientation Errors Sliding Mode Control

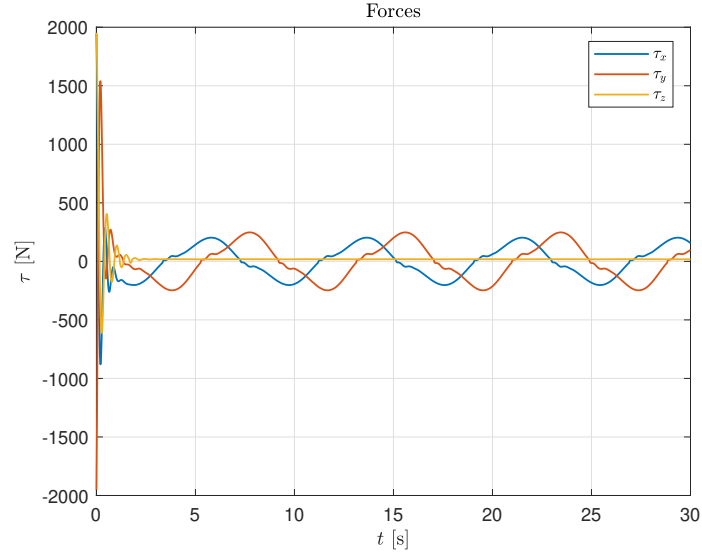


Figure 6: Forces Sliding Mode Control

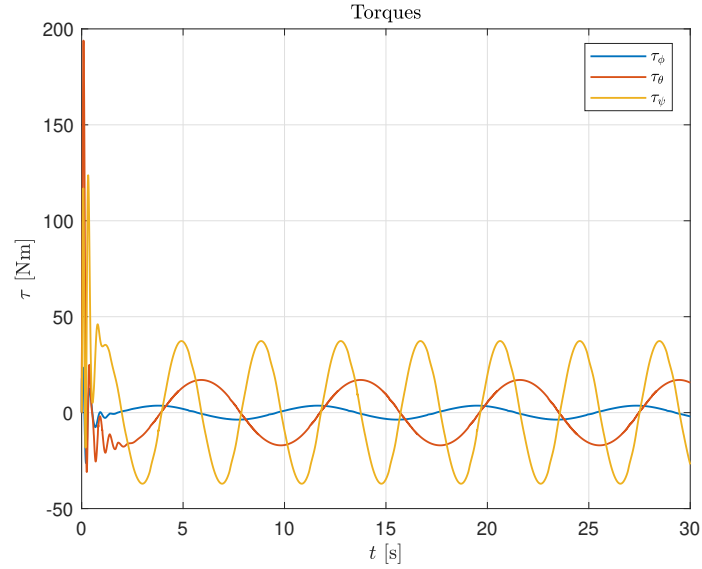


Figure 7: Torques Sliding Mode Control

From Fig.8-9 we can observe the linear and angular velocities. For the linear ones we can see that the heave component w is basically constant since the trajectory of the z -component is linear except at the start. Regarding the angular velocities, expect for an high initial value, they're close to zero for the rest of

the simulation.

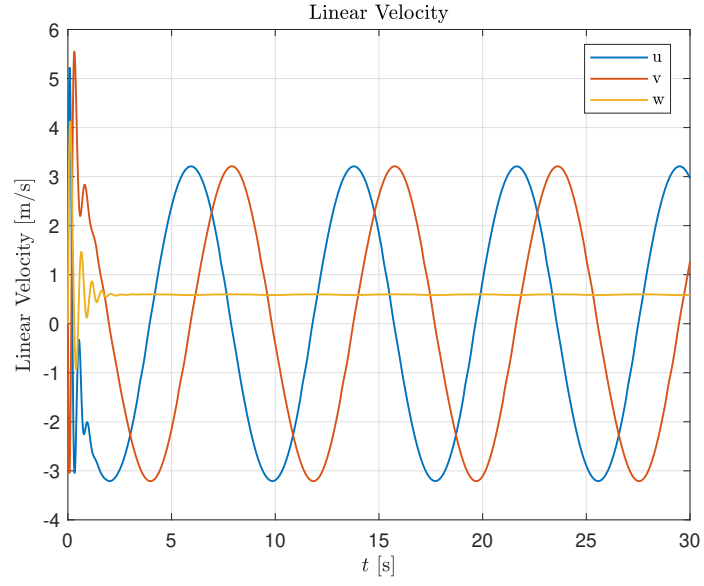


Figure 8: Linear Velocities Sliding Mode Control

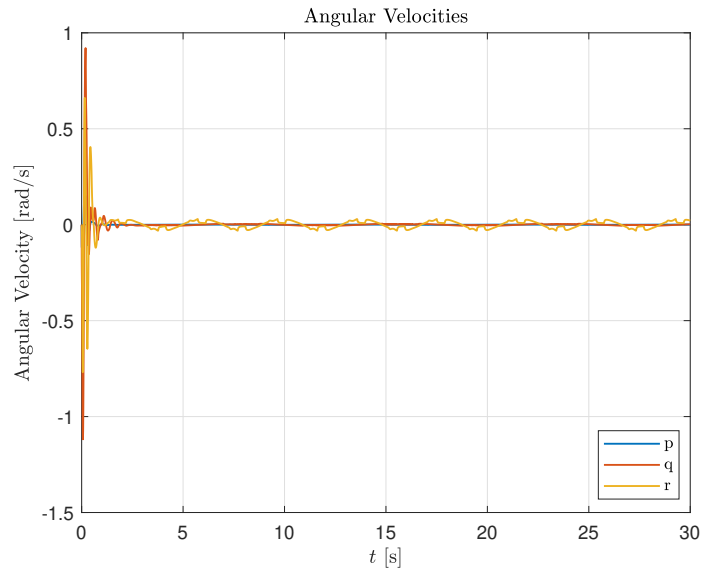


Figure 9: Angular Velocities Sliding Mode Control

10.2 Simulation Feedback Linearization

The gains of the commanded acceleration λ^n were selected as

- $K_D = 43$
- $K_P = 78$

In Fig. 10-11 we can observe the position and orientation errors. Also in this case the errors are norm-bounded, but the z component of the position error is higher than before (confront Fig.10 and Fig.4), while regarding the orientation error is about one order of magnitude larger with respect to the previous case (Fig.5-11).

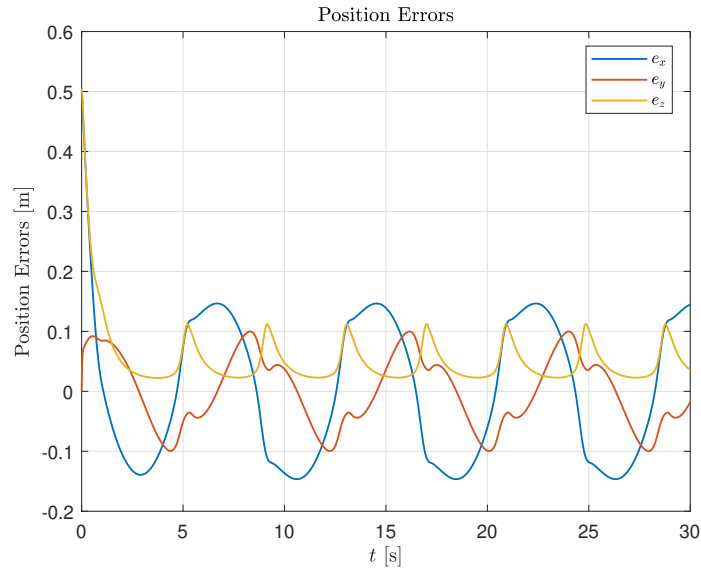


Figure 10: Positions Errors Feedback Linearization Control

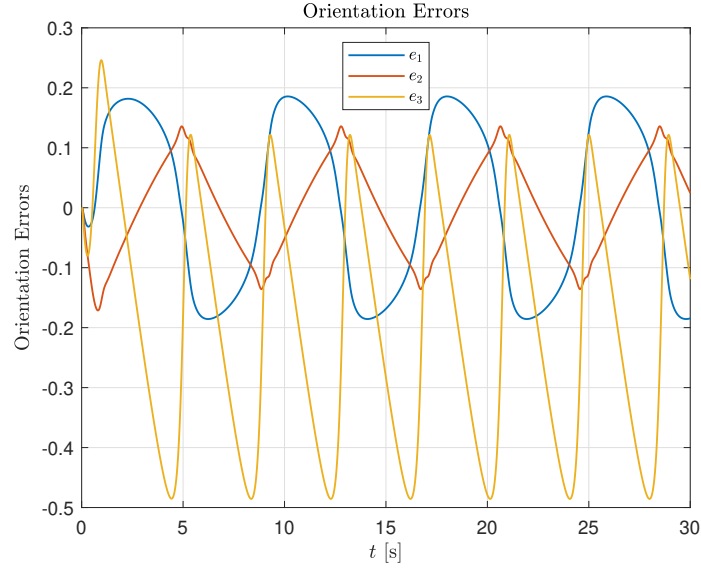


Figure 11: Orientation Errors Feedback Linearization Control

From Fig.12 we can see how the order of magnitude of the forces are the same as before, although the initial spikes are greater.

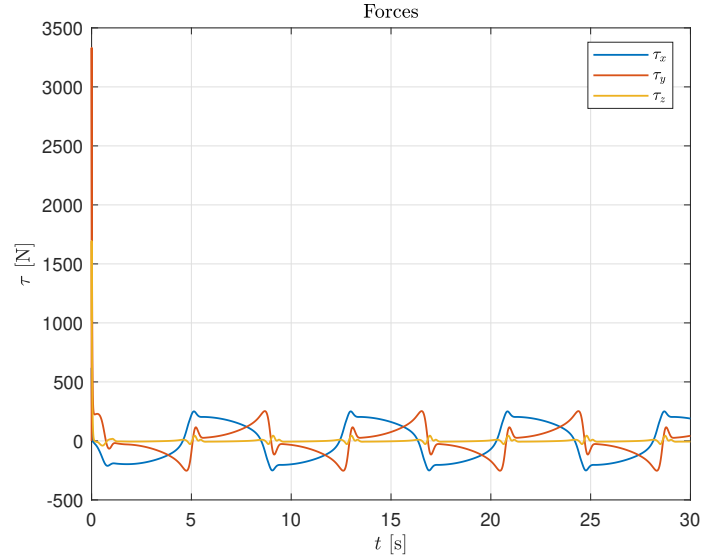


Figure 12: Forces Feedback Linearization Control

By comparing Fig.13 and Fig.7 the torques are less high than in the sliding controller case. Finally we can also infer that in the case of Feedback Lineariza-

tion Controller the control inputs have more high-frequency components than before: obviously if we hadn't used the boundary layer for the switching part of the sliding controller, the *chattering* effect would have been more evident and the control input would not have been so close to sinusoidal signals (Fig. 6-7).

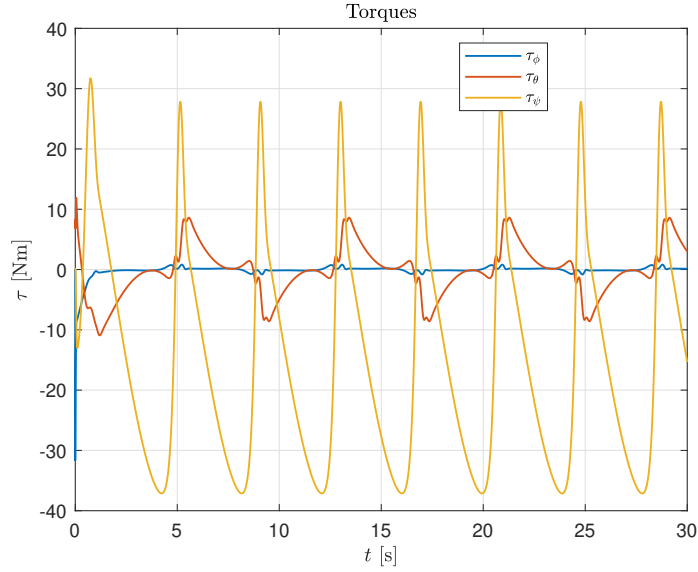


Figure 13: Torques Feedback Linearization Control

Finally, by comparing the angular and linear velocities (Fig.14-15) obtained with the preceding results (Fig.8-9), we can notice that in this case the maximum and minimum values of the linear velocities are smaller; on the other hand the angular velocities exhibit higher values than before, except for the yaw component r , which reaches similar values around zero as before.

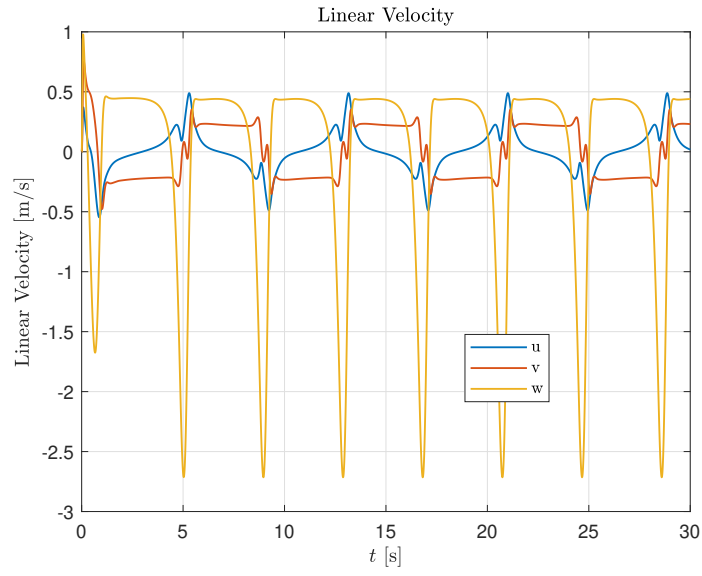


Figure 14: Linear Velocities Feedback Linearization Control

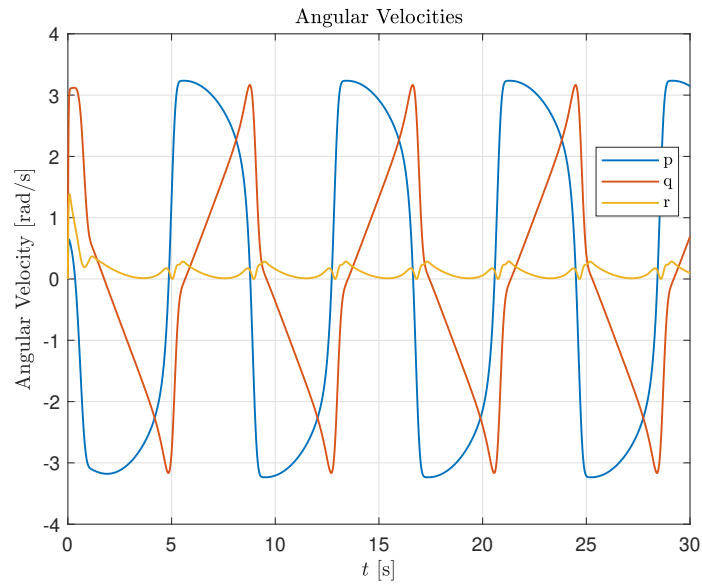


Figure 15: Angular Velocities Feedback Linearization Control

11 Conclusions

This project presented the modeling and control of the BlueROV2 Heavy using two nonlinear control strategies: sliding mode control and feedback linearization. The sliding mode controller demonstrated robust tracking performance with bounded errors. The feedback linearization approach simplified the system dynamics into a linear form, achieving satisfactory tracking but with slightly larger orientation errors. Both controllers successfully stabilized the ROV along a helical trajectory, although the sliding mode controller, once the *chattering* effect has been solved, had demonstrated better performances in term of tracking precision.

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